AIMLZG 549

API-driven Cloud Native Solutions

Assignment II

Group Details

Group No: <Fill Group No>

Group Member Names with Contribution

Sl. No	BITS ID	Name	Contribution of team member (Qualitative)	Percentage Contribution (Out of 100) Quantitative
1				
2				
3				
4				
5				
6				

General Instructions

- It is a Group Assignment.Each and every member of the group has to upload the files.
- You can use the same group as in assignment 1
- Weightage - 15 marks
- Please note that under no circumstances, the Assignment II deadline will be extended.
- **Submissions after the due date will result in deduction of marks [-1 per day].**

Project Details

I. I. Design and develop an API-based AI project that includes the following. [4 marks]

1. **Select a domain of your choice situated in a real-world context** (e.g., Education, Healthcare, Customer Service, Finance, etc.), where the AI solution can address a practical problem or workflow.

2. **Choose any one category** from the following:
    - Natural Language Processing (NLP)
    - Computer Vision (CV)
    - Speech Recognition
3. **Select at least three sub-tasks** across the chosen categories (e.g., Text Generation, Question Answering, Sentiment Analysis, Image Classification, Speech-to-Text, etc.).
4. **Identify appropriate models** (either Large Language Models (LLMs) or Small Language Models (SLMs)) to implement the selected sub-tasks.
5. **Design the solution such that all selected sub-tasks are cohesive**, contributing toward a **unified objective** in the chosen domain.
6. **Develop a small AI application** (Using Hugging Face, OpenAI, AWS Bedrock, or similar platforms).
7. The application must be **interactive and demonstrable** (e.g., Streamlit app, web interface, or API-based system).

II. II. Fine Tuning [2 marks]

- Select a relevant dataset from the chosen domain and fine-tune the model.

III. RAG Model [4 marks]

- **Implement a Retrieval-Augmented Generation (RAG) component**:
    - Use relevant domain-specific documents (e.g., PDFs, reports, knowledge bases, manuals, etc.).
    - Build a **chatbot interface** that can answer user queries using retrieved context.

IV. Situated Learning – Workplace Application [5 marks]

Apply the AI solution developed in Sections I–III to a practical problem, process, or workflow in the your workplace or relevant industry. You may use anonymised or synthetic data where workplace confidentiality prevents use of real data.

1. Anchor – Identify a Workplace Problem [1 mark]

Identify a relevant workplace system, dataset, process, or problem. Describe the organisational context, current workflow, limitations, and expected benefit of the proposed AI solution. If a workplace artifact is unavailable, use a close analogue from the same industry.

2. Apply – Adapt the AI Solution to the Workplace [2 marks]

Identify suitable workplace data sources, documents, APIs, or knowledge bases. Explain and demonstrate how the chosen AI sub-tasks, fine-tuned model, and RAG chatbot support the identified workflow using representative inputs and outputs.

3. Analyse – Evaluate Workplace Relevance [1 mark]

Evaluate the demonstrated outputs against workplace needs (e.g., response quality, accuracy,

efficiency, scalability). Explain the gap between the prototype and what the organisation would require for practical use.

4. Reflect – Deployment Constraints and Improvements [1 mark]

Discuss constraints such as data privacy, security, regulation, infrastructure, cost, and integration with existing systems. Identify and justify the single most important improvement required before workplace deployment.

Situated Learning marking rubric: Anchor 1 + Apply 2 + Analyse 1 + Reflect 1 = 5 marks.

Overall assessment: I. API-based AI Application 4 + II. Fine-Tuning 2 + III. RAG Model 4 + IV. Situated Learning 5 = 15 marks.

The report must include a separate "Situated Learning – Workplace Application" section with the workplace context, evidence of application, analysis, and reflection.

Evaluation Guidelines

Prepare a word or pdf document containing the project details (screenshots of the application are required, along with explanation). Highlight clearly the contribution of each group member in executing the assignment. Upload the file as <groupid.docx> or <groupid.pdf> into the portal. Note that each student needs to upload the same file individually**(except the Situated Learning – Part 4).**

Prepare a video demonstrating the entire project. Upload the Video into the google drive. Submission of both the Word/PDF document and demonstration video is mandatory. Submission quality and evidence are assessed within the 15-mark component rubric; no additional marks are allocated.

Note

1. No extension of deadline will be given under any circumstances.
2. Since Assignment-II is part of EC-1 component, there is NO MAKEUP for this component.